



Air University
Department of Cyber Security
(Final-Term Examination: Fall 2025)

Student ID: _____

Student Sign: _____

Subject: Computer Organization and Assembly Language

Class: BS-Cyber Security

Code: CS-226

Section: A B

FM Name: Ms. Maryam Malik

FM Signature: _____

Total Marks: 100

Date: _____

Max Duration: 3 Hours

HoD Signature: _____

Instructions:

- You are required to attempt **ALL Questions**.
- This is a closed book/notes exam.
- Calculators are allowed,
- **Return question paper with the answer sheet**

Q. No	Questions	CLO	Marks
1	A. Differentiate between the Local and Global Labels with example. (5 marks) B. Describe how the XOR instruction can be used to toggle the case of an ASCII character stored in the AL register. (5 marks) C. How does the DIV instruction perform unsigned division? Which registers hold: dividend, quotient. Remainder for 8,16 and 32-bit register. (5 marks) D. What values are returned by TYPE, LENGTHOF, and SIZEOF for the following? <i>array WORD 10,20,30,40</i> E. Explain the role of the CALL instruction in procedure execution. What does the RET instruction do internally with respect to the stack? (5 marks) F. <i>.data</i> <i>num1 WORD 1234h</i> <i>num2 WORD 0100h</i> <i>result DWORD ?</i> <i>.code</i> <i>mov ax, num1</i> <i>mul num2</i> Write the required assembly instructions to store the complete 32-bit multiplication result (DX:AX) into the single variable result.	1	30

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- A. Given the following assembly code, determine the values in the registers after each instruction is executed.

```
varX BYTE 05h, 15h, 25h, 35h, 45h, 55h
varY WORD 0A1Bh, 2C3Dh
varZ DWORD 11223344h, AABBCDDh
```

```
mov cx, WORD PTR [varX+1] ; CX = ?
mov dl, BYTE PTR [varZ+6] ; DL = ?
mov ax, WORD PTR [varY+2] ; AX = ?
mov eax, DWORD PTR [varZ+4] ; EAX = ?
mov bx, WORD PTR [varX+3] ; BX = ?
```

- B. Use .WHILE directive to print numbers from 1 to 10.
 C. Write a program that turns a lowercase character into uppercase using bitwise instructions only (no arithmetic).
 D. An array of signed words is given. Write a program to find the first positive number in the array using LOOPNZ.

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- A. In the following instruction sequence, show the values of the Carry, Zero, and Sign flags where indicated:

```
mov al, 00001111b
test al, 00000010b ; a. CF= ZF= SF=
mov al, 00000110b
cmp al, 00000101b ; b. CF= ZF= SF=
mov al, 00000101b
cmp al, 00000111b ; c. CF= ZF= SF=
```

- B. Write an assembly program to **encrypt** a string stored in memory using a single-byte encryption key and decrypt it back to its original form. The string, "SECURE", is stored in the .data section using key (5Ah). Your program should process each character in a loop, stop at the null terminator, and overwrite the original message with the encrypted and decrypted results.
- C. What will be the value of BX after the following instructions execute? (10 marks)
- ```
mov bx, 029D6h
xor bx, 8181h
```

\*\*\*\*\*End of Paper\*\*\*\*\*